

What Is the User Experience of Eyes-Free Touch Input with Vibrotactile Feedback Decoupled from the Touchscreen?

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ABSTRACT

Vibrotactile cues commonly serve as substitutes for visual feedback in *eyes-free* touch input, yet not all touchscreens have haptic capabilities. In this work, we examine the user experience of arm-localized vibrations delivered away from the touchscreen through a connected wearable. In a controlled experiment, fourteen participants used a touchscreen in *eyes-free* mode, while receiving single-pulse vibrations on the index finger, wrist, or forearm as confirmatory feedback of their on-screen touches. We reveal a favorable user experience of vibrations decoupled from the touchscreen, characterized by high perceived enjoyment (Mdn=4 on a scale from 1 to 5), efficiency (Mdn=4), integration (Mdn=5), and confidence (Mdn=4) with moderate complexity (Mdn=3) and low perceived distractedness (Mdn=2), difficulty (Mdn=1), and confusion (Mdn=1). To contextualize our findings, we contrast the user experience of arm-localized vibrations in *eyes-free* vs. *eyes-on* touch input, and outline preliminary observations involving three legally blind users.

CCS CONCEPTS

• **Human-centered computing** → **Touch screens; Haptic devices; Gestural input.**

KEYWORDS

User experience, eyes-free input, vibrotactile feedback, wearables, touch input, touchscreens

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1 INTRODUCTION

Vibrotactile feedback is commonly used in mobile interactive systems for notification delivery [2,33], navigation assistance [18,21], text entry [5], user authentication [14], tangible computing [31], cross-device interactions [26], confirmatory feedback for mid-air input [23], and generic manipulation of digital content [36]. Moreover, prior work has shown that vibrotactile feedback enhances the user experience of touchscreen interaction [27], including in contexts of use characterized by visual impairments [9,24,25], by supplementing or compensating visual with haptic cues.

However, not all touchscreen devices have built-in capabilities for vibrotactile feedback [13,27], while those that natively support it, such as smartphones, are occasionally used in *eyes-free* mode when the user's visual attention cannot be directed to the device itself [11,34]. For example, touch input is frequently performed eyes-free during activities such as walking, jogging, driving, or in situations involving encumbrance to prevent potential hazardous or dangerous outcomes caused by carelessly smartphone usage when visual attention must focus on another task. In this context, vibrations delivered away from the device and, hence, distal to the touch point, can provide haptic cues to compensate either natively missing haptics or the visual cues primarily provided by the touchscreen device. Examples include on-finger vibrations delivered by an electronic ring for surface computing [13] and on-wrist haptics, delivered by a smartwatch, when biking or driving [4]. However, prior research in this area has primarily focused on aspects of user efficiency and performance [5,10,16,20], whereas the user experience (UX) of distal, arm-localized haptics has been largely overlooked despite the advantages it brings for cross-device interaction where input on one device is complemented with vibrotactile feedback on another, worn on the arm implementing touch input.

In this paper, we contribute empirical results from a controlled experiment conducted with N=14 participants that rated their experience of arm-localized vibrations during *eyes-free* input with a touchscreen device, which we evaluated across eight distinct UX dimensions—perceived enjoyment, efficiency, integration, confidence, distractedness, complexity, difficulty, and confusion of the cross-device interaction. Our findings reveal a favorable user experience of vibrotactile feedback at the *finger*, *wrist*, and *forearm* levels

compared to touch input without confirmatory vibrations. Furthermore, unlike prior findings obtained for arm-localized vibrotactile feedback in *eyes-on* contexts of use [10,28], we did not detect any difference between the experience of vibrations on various parts of the arm when users were not looking at the touchscreen. The highly favorable experience of arm-localized vibrations in these findings suggests that users primarily appreciate the benefits of vibrotactile feedback regardless of where feedback is delivered on the arm during eyes-free input. While our primary target audience is users without visual impairments who employ touchscreen devices in eyes-free mode, we also present preliminary results regarding the experience of three legally blind users, revealing a few differences. We capitalize on our findings to suggest future work in distal vibrotactile feedback accompanying eyes-free touch input.

2 RELATED WORK

We relate to prior work on vibrotactile feedback delivered on the body with the aim to enhance the user experience of interacting with touchscreen devices. Most of this work has focused on technical, engineering aspects of implementing vibrotactile feedback [10,19,22,28]. For example, Henderson *et al.* [10] examined the effect of vibrations delivered through a smartwatch on the user's wrist on the error rate and task time of smartphone interactions. Forlines *et al.* [7] reported that vibrotactile feedback delivered via a haptic stylus, even in the form of subtle "soft click"-like vibrations, increases user performance during pointing and crossing selection tasks and leads to shorter selection times. McAdam and Brewster [17] discovered that distal vibrations on the upper arm significantly increase text entry rates on a tabletop computer, without compromising accuracy, compared to the baseline condition of no vibrotactile feedback. Liao *et al.* [15] introduced Dwell+, a technique that enables multi-level dwell selection using rapid haptic ticks, applicable to smartphones, smartwatches, styli, and HMDs.

In this space, very few papers have examined the user experience of on-body haptics in relation to touchscreen input. Terenti and Vatavu [27] presented empirical results on the user experience of vibrotactile feedback delivered on the finger, wrist, and forearm during interactions with large touchscreen displays, and reported participants' significant preference for vibrations on the finger. Gaudeni *et al.* [8] evaluated vibrotactile feedback of various intensities and frequencies delivered by a haptic ring on the fingerpad, distal phalanx of the index finger, and wrist. Vatavu *et al.* [31] used vibrotactile feedback on the index finger to induce the illusory sensation of holding virtual content between one's fingers after picking it up from a smartphone's touchscreen. They reported a user experience characterized by high perceived enjoyment and credible feeling of presence of the invisible virtual objects situated in one's hand, with moderate distractedness—all hallmarks of the user experience resulting from non-natural interaction design [30].

This prior work has shown that vibrotactile feedback delivered distally to the location of the finger touching the screen leads to both increased user performance and better user experience. However, these findings have systematically centered on *eyes-on* touch input, whereas the user experience of vibrotactile feedback delivered away from the screen during *eyes-free* input has been overlooked so far. In this work, we present the first insights into users' experience of

vibrations delivered at the *finger*, *wrist*, and *forearm* level through a wearable device, intended to serve as confirmatory feedback of touch input when users do not or cannot look at the touchscreen. In the next section, we present our experiment designed to evaluate specific elements of this novel user experience.

3 EXPERIMENT

We conducted a controlled experiment to evaluate the user experience of vibrotactile feedback accompanying eyes-free touch input, delivered away from the touchscreen at various locations on the arm—*finger*, *wrist*, and *forearm*,—progressively more distant from the touch point. The experiment was conducted at Istituto Italiano di Tecnologia, Genova and the University of Suceava and followed the ethical procedures in place at each institution.

3.1 Participants

We recruited N=14 participants through convenience sampling, aged between 19 and 34 years old ($M=22.7$, $SD=3.8$), of which six self-identified as males and eight as females. All participants were right-handed. Eleven of the participants (78.6%) reported using PCs, six (42.9%) tablets, five (35.7%) smart earbuds, and five (35.7%) smartwatches or fitness trackers. Furthermore, all of the participants were regular smartphone users, of which nine (64.3%) reported having the keyboard vibration feature turned on on their smartphones.

3.2 Apparatus

We developed a mobile video game intended to be played in eyes-free mode. The game splits the display of the mobile device into two panels, left and right, where touching one of the panels increases the score and touching the other decreases it. The effect of each panel is randomly assigned before each trial/touch and announced with audio feedback, e.g., "Click on the left panel." When touching the indicated panel, a cheerful sound would play, while touching the other panel and, thus, losing a point in the game, would result in a strident alert sound. Moreover, a custom-built wearable device, controlled via an Arduino Nano microcontroller, delivers vibrotactile feedback on the user's arm. The vibrations are generated by three 10mm DC (ERM) coin vibration motors¹ incorporated into a flexible body with adjustable length and Velcro straps, which enable easy donning of the motors onto the participant's index finger, wrist, and forearm; see Figure 1 for photographs taken during the experiment, including a close up of our wearable device. Vibration design was simple, consisting of a 150ms-long pulse, a value determined empirically during technical prototyping so that vibrations could be felt unambiguously on the three arm regions of different sensibilities. Vibration intensity was controlled through pulse-width modulation. We deployed the application on a Samsung Tab A tablet (1.2GHz Quad-Core CPU, 1.5GB RAM); see Figure 1.

3.3 Design

We designed our experiment following a within-subject repeated-measures protocol with two independent variables:

- BODY-LOCATION, nominal variable with four conditions, *finger*, *wrist*, *forearm*, and *none*, where the latter represents

¹<https://nfpshop.com/product/10mm-coin-vibration-motor-3mm-type-model-nfp-c1030>

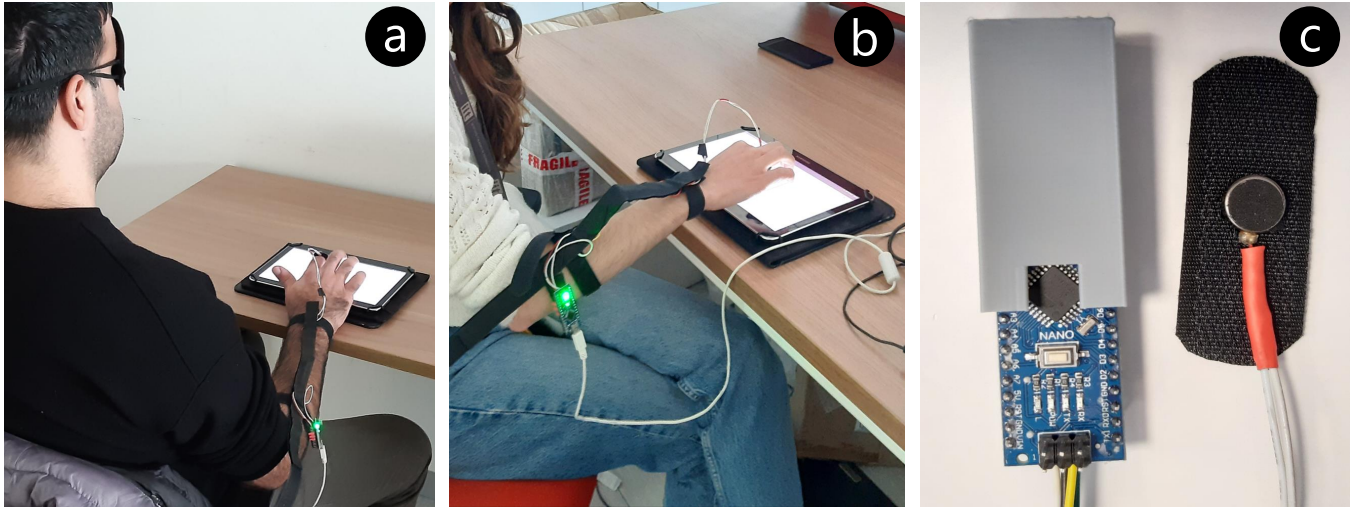


Figure 1: Photographs of two participants in the eyes-free touch input experiment (a, b) and a close up of our wearable device (c) designed for arm-localized vibrotactile feedback, showing the Arduino board and one of the three vibration motors.

the control baseline of no vibrotactile feedback. The three locations on the arm where vibrations were delivered were specifically chosen to be progressively distant from the touch point and in direct correspondence to locations on the arm where a ring, watch, and armband would be worn.

- **INTENSITY**, ordinal variable with two conditions, *default* and *custom*. In the *default* condition, adopted from [27], the vibration motors were powered at the maximum permissible voltage, which was 5V in our case. The *custom* condition involved determining specific intensity levels per participant so that vibrations on the *finger*, *wrist*, and *forearm* would be comfortably perceived as having equal intensity given subjective differences between participants and different sensibilities of the arm regions; see Subsection 3.4 for details.

The dependent variables are represented by specific UX measures, described in detail in Subsection 3.5.

3.4 Task

After signing a consent form, participants were presented the goal of the experiment and engaged in a training stage to become accustomed with the game and vibrotactile feedback delivered by our wearable device. The experiment consisted of two parts, encompassing all combinations of **BODY-LOCATION** and **INTENSITY** conditions, during which participants played the game in eyes-free mode. Touch input was always performed with the index finger. Some of the participants used a blindfold, which we provided, while others found more comfortable to keep their eyes shut. Due to the novelty of distal haptics [10,27], we always administered the *default* intensity condition first, so that the participants would have a baseline to refer to before engaging with the more nuanced condition of *custom* vibration intensity levels per finger, wrist, and forearm. In the second part of the experiment, we used the staircase method [12] to adjust the vibration intensities for each of the three arm-level locations so that vibrations would be subjectively

perceived of equal intensity. The location where vibrations were perceived the weakest was used as the baseline (or standard) during the staircase procedure. Two staircases corresponding to the other two locations were derived to equalize the perceived intensity of vibrations delivered by our wearable.² Subsequently, we applied the same procedure from the first part of the experiment to evaluate the user experience of vibrotactile feedback during eyes-free touch input. The order of **BODY-LOCATION** was randomized across participants for each **INTENSITY** condition. In total, each participant engaged in 50 trials for each combination of **BODY-LOCATION** × **INTENSITY**. The experiment lasted 45 minutes per participant.

3.5 Measures

We administered a questionnaire to collect information about our participants' user experience of vibrotactile feedback delivered at various locations on the arm during eyes-free touch input. The questionnaire consisted of the Usability Metric for User Experience (UMUX) [6], intended as an overall gauge of our participants' interaction experience, as well as eight specific UX measures, adopted from [27]. The latter were designed to evaluate aspects of the user experience of vibrotactile feedback with respect to the control condition of no vibrations (*none*) using both positively and negatively connoted descriptors, as follows:

- **PERCEIVED-ENJOYMENT**, measured on a 5-point Likert scale from 1 (strongly disagree) to 5 (strongly agree) in response to the statement "Interacting with the display felt more enjoyable with vibrotactile feedback than without."
- **PERCEIVED-EFFICIENCY**, "Interacting with the display felt more efficient with vibrotactile feedback than without."
- **PERCEIVED-CONFIDENCE**, "Vibrotactile feedback made me feel more confident when interacting with the display compared to when vibrations were absent."

²Following the staircase procedure, we observed that the vibration intensity on the *finger* decreased to an average of 88.4% (SD=11.6%) of the maximum intensity. The intensity also decreased on the *wrist* to 95.0% (SD=7.6%) and *forearm* to 98.3% (SD=2.3%).

- PERCEIVED-INTEGRATION, “Touching the display and feeling the vibrations integrated well into one experience.”
- PERCEIVED-DISTRACTENESS, measured in response to the statement “Interacting with the display felt more distracting with vibrotactile feedback than without.”
- PERCEIVED-DIFFICULTY, “Interacting with the display felt more difficult with vibrotactile feedback than without.”
- PERCEIVED-CONFUSION, “Vibrotactile feedback created confusion for me when interacting with the display.”
- PERCEIVED-COMPLEXITY, “Vibrotactile feedback was unnecessarily complex when interacting with the display.”

Given the ordinal nature of these variables, we report medians (Mdn) as the preferred measure of central tendency accompanied by means (M) for comprehensiveness, and employ ANOVA with the Aligned-Rank Transform [35] for nonparametric factorial analyses of the effects of BODY-LOCATION and INTENSITY.

4 RESULTS

The results of the UMUX test revealed an overall favorable experience of vibrotactile feedback during eyes-free touch input for both *default* (Mdn=95.8, M=91.4) and *custom* (Mdn=91.7, M=89.9) vibration intensities, measured on a scale from 0 (low) to 100 (maximum). Despite a slightly higher (+4.5%) average score observed in the *default* intensity condition, there was no statistically significant effect of INTENSITY on UMUX ($F_{(1,13)}=0.096$, $p=.761$, $n.s.$).

However, we found that INTENSITY affected participants’ ratings of three of our eight specific UX measures: PERCEIVED-ENJOYMENT ($F_{(1,65)}=10.460$, $p=.002$, $\eta_p^2=.14$), PERCEIVED-CONFIDENCE ($F_{(1,65)}=14.927$, $p<.001$, $\eta_p^2=.19$), and PERCEIVED-DISTRACTENESS ($F_{(1,65)}=4.381$, $p=.040$, $\eta_p^2=.06$), respectively. This finding indicates that our participants enjoyed more (Mdn=4.5, M=4.2 vs. Mdn=4.0, M=3.4), were more confident (Mdn=4.0, M=4.1 vs. Mdn=4.0, M=3.4), and less distracted (Mdn=1.0, M=1.9 vs. Mdn=2.0, M=2.0) by vibrations for which the intensity was *customized* per participant, finger, wrist, and forearm; see Figure 2. However, we did not observe statistically significant effects of INTENSITY on PERCEIVED-EFFICIENCY, PERCEIVED-INTEGRATION, PERCEIVED-CONFUSION, or PERCEIVED-COMPLEXITY (all p -values $>.05$, $n.s.$).

We also found that the perceived user experience of vibrotactile feedback during eyes-free touch input was overall similar across the three arm locations—*finger*, *wrist*, and *forearm*,—with statistically significant differences identified only for the PERCEIVED-CONFIDENCE measure ($F_{(2,65)}=9.088$, $p<.001$, $\eta_p^2=.22$). According to this finding, our participants felt more confident during eyes-free touch input when vibrations were delivered on the *finger* (Mdn=4.5, M=4.2) compared to the *wrist* (Mdn=4.0, M=3.6, $p=.039$) and the *forearm* (Mdn=3.5, M=3.4, $p<.001$), respectively (Bonferroni corrections applied). We also detected a significant interaction between INTENSITY and BODY-LOCATION for this measure ($F_{(2,65)}=3.570$, $p=.034$). The rest of our specific UX measures revealed favorable perceptions of vibrotactile feedback during eyes-free touch input compared to the baseline condition of no vibrations (*none*), yet not discernible ($p>.05$, $n.s.$) across the specific regions of the arm examined in our experiment. We found high PERCEIVED-ENJOYMENT (Mdn=4.0, M=4.0), high PERCEIVED-EFFICIENCY (Mdn=4.0, M=3.9),

high PERCEIVED-INTEGRATION (Mdn=5.0, M=4.4), low PERCEIVED-DISTRACTENESS (Mdn=2.0, M=1.9), low PERCEIVED-DIFFICULTY (Mdn=1.0, M=1.9), low PERCEIVED-CONFUSION (Mdn=1.0, M=1.7), and moderate PERCEIVED-COMPLEXITY (Mdn=3.0, M=2.9). Figure 2, bottom right shows Spearman correlation coefficients computed for the eight UX measures in the *default* (upper matrix) and *custom* (lower matrix) conditions, with significant correlations highlighted (Bonferroni control of Type I error rate at $\alpha=.05/(8\cdot7/2)=.00179$). Aside a few exceptions in the *custom* condition, these findings show that our measures reflect distinct facets of distal vibrations UX.

5 DISCUSSION

We found that the UX of arm-localized vibrotactile feedback during eyes-free touch input was favorably rated with no significant effect of the location where vibrations were delivered, from *finger* to *forearm*—results corroborated by multiple measures involving both positively and negatively connoted descriptors.

Our findings have implications for eyes-free interaction, opening up interesting directions for future work. First, they point to a technical solution for touchscreens without native vibrotactile capabilities by leveraging external vibrations delivered by a connected device worn on the arm performing touch input. The latter may include a smartwatch or electronic ring employed in tandem with the touchscreen for confirmatory feedback upon on-screen touches. Such confirmation is especially important to increase users’ confidence, albeit lack of visual cues, that input was correctly registered by the touchscreen. Second, our findings show that user confidence can be further enhanced through user-dependent customization of vibration intensity. In fact, PERCEIVED-CONFIDENCE was among the three UX measures, alongside PERCEIVED-ENJOYMENT and PERCEIVED-DISTRACTENESS, that were significantly affected by INTENSITY, with user-dependent intensity levels outperforming the default one. Noteworthy, high confidence was complemented by high enjoyment and low distractedness during eyes-free input.

To put these results into a broader context and offer more opportunities for further work in this space, we discuss them in relation to (i) prior findings from the scientific literature on the UX of arm-localized vibrations for *eyes-on* touch input (Subsection 5.1) and (ii) the reported experience of three legally blind individuals, which we additionally recruited for a follow-up investigation (Subsection 5.2).

5.1 The UX of Vibrotactile Feedback in Eyes-Free vs. Eyes-On Touch Input

To our knowledge, Terenti and Vatavu [27] reported the only results available on the UX of vibrotactile feedback at *finger*, *wrist*, and *forearm* level during touch input, but in an *eyes-on* context of use. Since we designed our experiment following theirs in terms of the BODY-LOCATION conditions (see Subsection 3.3) and both sample sizes are equal ($N=14$ participants), we discuss our results on *eyes-free* touch input with respect to the new control condition represented by *eyes-on* input by leveraging the dataset from [27].³ Given that the setups of the two experiments differ, it is important to regard this comparison as preliminary in nature, yet we believe informative to foster future work. Figure 3 presents histograms of our participants’ ratings of the eight specific UX measures mirrored

³The data collected in [27] is available from <https://doi.org/10.5281/zenodo.6642532>.

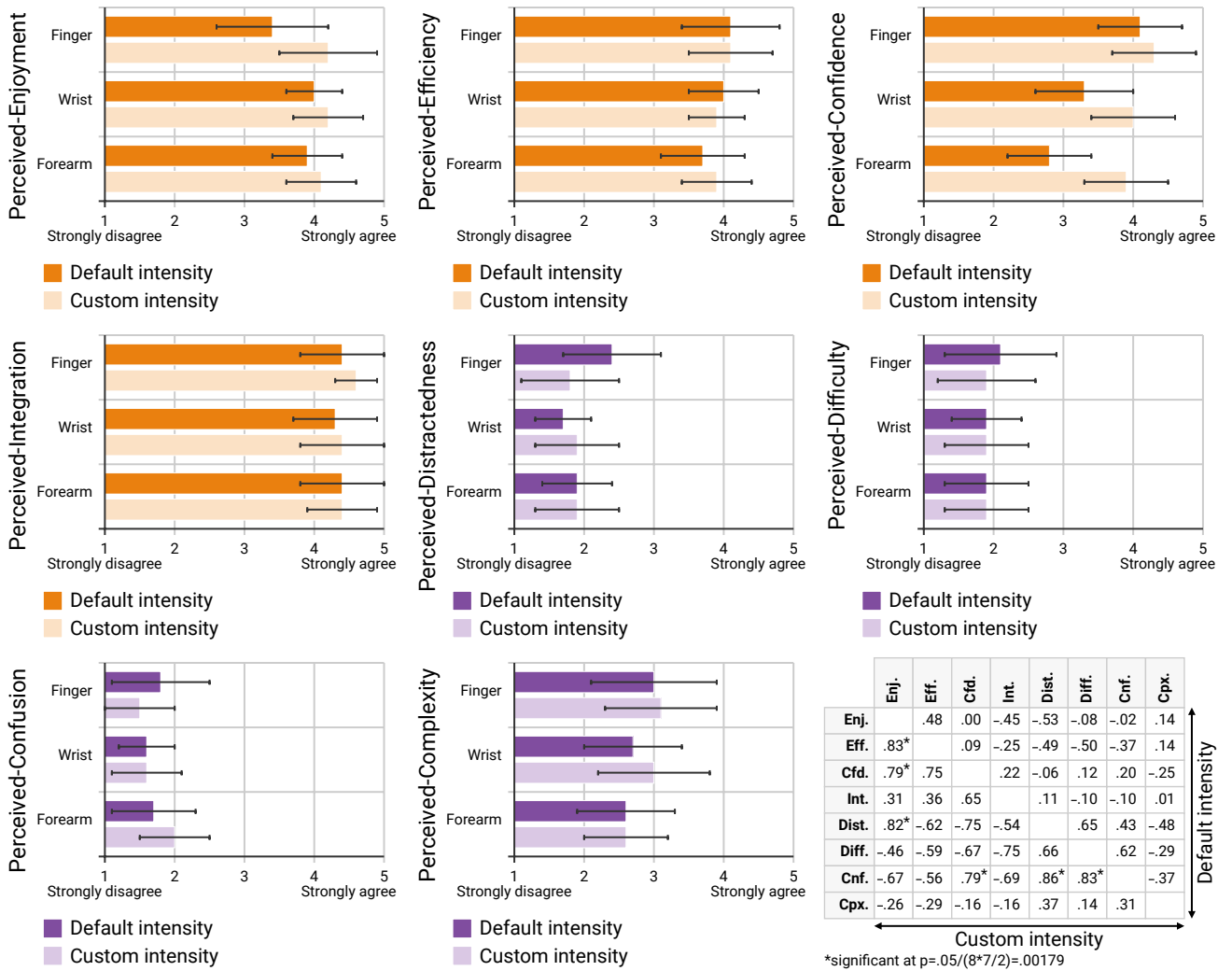


Figure 2: The UX of vibrotactile feedback on the *finger*, *wrist*, and *forearm*, relative to the *none* baseline, during eyes-free touch input. *Note*: higher scores show better UX for the positively connoted measures (orange), while lower scores indicate better UX for the negatively connoted ones (violet). The bottom right table shows Spearman correlation coefficients (N=14).

by those reported by the fourteen participants in [27]. One immediate observation is that vibrotactile feedback is preferred to no vibrations in both *eyes-free* and *eyes-on* input due to considerably more ratings above 3 for the positively connoted UX measures and, respectively, below 3 for the negatively connoted ones. Figure 3, bottom right shows Spearman correlation coefficients for the UX measures in the *eyes-on* (upper matrix) and *eyes-free* (lower matrix) experiments, aggregated by BODY-LOCATION. None of the correlations were statistically significant (Bonferroni control of Type I error rate at $\alpha=.05/(8 \cdot 7/2) = .00179$), and most (82.1%) were medium ($<.5$) or small ($<.3$), a result that confirms the usefulness of our set of measures to characterize the multi-faceted UX of distal vibrotactile feedback. A second observation, revealed by Kruskal-Wallis

rank sum tests,⁴ is that the UX of arm-localized vibrotactile feedback is not significantly affected (p -values $>.05$) by input mode. Although this finding indicates that arm-localized vibrations are equally useful in both *eyes-free* and *eyes-on* touch input, we recommend future examination with a rigorously controlled experiment. We also recommend replicating our findings using off-the-shelf wearables, such as smartwatches and rings, for readily-available distal vibrotactile feedback during touch input.

⁴We used Kruskal-Wallis rank sum tests to compare the two datasets, corresponding to a between-subject experiment design involving one independent factor with two conditions, i.e., *eyes-free* and *eyes-on* touch input.

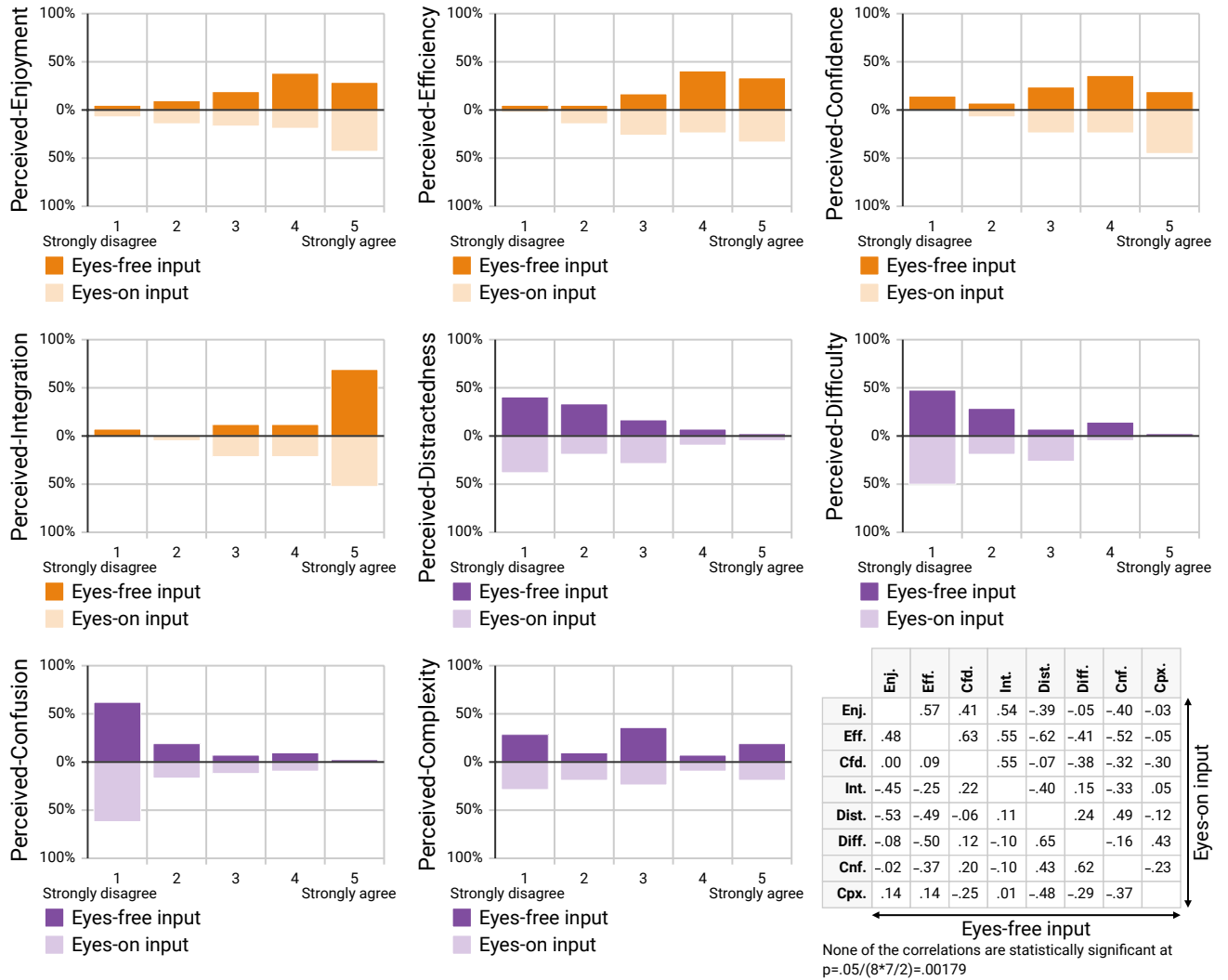


Figure 3: The UX of vibrotactile feedback during *eyes-free* (our experiment) and *eyes-on* (the experiment in [27]) touch input. **Note:** higher scores show better UX for the positively connoted measures (orange), while lower scores indicate better UX for the negatively connoted ones (violet). The bottom right table shows Spearman correlation coefficients (N=14).

5.2 Insights From Blind Users

Previous studies [32] have shown that blind people display higher vibrotactile discrimination abilities, which could be leveraged for more accessible touchscreen interaction design [29]. To gain insight into whether blindness impacts our findings, we recruited three legally blind participants through an internal database of people with visual impairments: P₁ (male, 35 years old, degenerative oculopathy), P₂ (female, 57 years old, retinitis pigmentosa), and P₃ (male, 64 years old, uveitis). They engaged in the same experimental task as described in Subsection 3.4 and were paid 20 euros. The experiment lasted 60 minutes per participant.

We found that the average UMUX score was 10.8% higher in the *custom* intensity condition (84.7) compared to *default* (76.4), in

contrast to the results obtained with sighted participants where the situation was reversed (91.7 vs. 95.8). This finding shows the need for customizing vibrotactile feedback to leverage the higher tactile sensitiveness for cutaneous input found in blind compared to sighted individuals [1]. We also found the UX of *custom* vibrations characterized by highly perceived integration with touch input (Mdn=5), high enjoyment (Mdn=4), efficiency (Mdn=4), and confidence (Mdn=4) as well as low distractedness (Mdn=2), difficulty (Mdn=1), confusion (Mdn=1), and complexity (Mdn=1). While these preliminary results indicate a nuanced UX compared to that reported by sighted users, we recommend conducting a between-subject experiment with a larger sample to rigorously examine the nature of such UX differences. Future work may also look at

combining vibrations with other haptic modalities, such as kinesthetic [3] or thermal [37], to further leverage specialized tactile discrimination in users with various visual abilities.

5.3 Limitations

Our experiment has a few limitations. First, we did not randomize the conditions of the INTENSITY independent variable, and always administered the *default* intensity first so that participants would have a baseline to refer to before engaging with the more complex *custom* intensities at various arm locations. Second, the *none* condition was always presented first for the same reason. Replication experiments can be conducted to address these limitations.

6 CONCLUSION

We examined the user experience of arm-localized vibrotactile feedback during *eyes-free* input with touchscreen devices, and reported favorable ratings that complement prior results obtained for *eyes-on* input. Furthermore, we reported preliminary results on the user experience of three legally blind individuals with the same setup. Since distal vibrations can be readily delivered through commercially available wearable devices, such as smartwatches, rings, and armbands, interesting opportunities arise for cross-device interaction to enhance the experience of touchscreen input with mobile devices for various user groups and in various contexts of use.

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